

RODRIGO LAIA

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EDUCATION

MSc in Computer Science and Engineering (Artificial Intelligence)

Instituto Superior Técnico

 Oct 2024 – Present

- Average: **18/20** (Portugal: **0–20**, 20 best; 10 passing).
- Received Técnico's **Academic Merit Award (2024/2025)**.
- Coursework: NLP, Deep Learning, Reinforcement Learning, Search & Planning.

BSc in Physics and Technological Engineering

Instituto Superior Técnico

 Oct 2021 – Sep 2024

- Average: **16/20** (Portugal: **0–20**, 20 best; 10 passing).
- BSc project: *Machine Learning as a Tool to Improve Nuclear Fusion Experiments* (resulted in a peer-reviewed publication).
- Presentation: FÍSICA 2024 (24th National Physics Conference & 34th Iberian Meeting for Physics Education).

Athens Program

Czech Technical University

 Nov 2023

- Introduction to Lean Six Sigma Management.

RESEARCH EXPERIENCE

Researcher

INESC-ID / Instituto Superior Técnico

 Sep 2025 – Present  Lisbon, Portugal

- Currently conducting my master's thesis on **translation and classification between European and Brazilian Portuguese**, under the supervision of Professor Arlindo Oliveira and Professor Bruno Martins.

Researcher (Machine Learning)

Diffusion Dynamics

 Nov 2024 – Mar 2025  Hybrid

- Designed a time-series **anomaly detection** pipeline for a **Tier-1 network provider** to flag persistent outages from high-volume events.
- Built scalable pipelines (SQLite + batch processing) and shipped an evaluation dashboard + **Gradio** explorer to speed analyst workflows.
- Defined an evaluation protocol prioritizing actionable incidents while reducing transient-noise false alarms.

Research Collaborator

University of Wisconsin–Madison

 Jan 2024 – Dec 2025  Remote

- Released a dataset of **12.4M** stellarator configurations with feasibility metrics for fusion research.
- Achieved **95% accuracy** distinguishing feasible stellarators by optimizing **LightGBM** models.
- Built reproducible Python experiment pipelines and evaluation for large-scale physics simulations.

Research Intern

LIP (Laboratório de Instrumentação e Física Experimental de Partículas)

 Jun 2022 – Sep 2022

- Built a high-performance **Unity/C#** visualization tool integrating Monte Carlo simulations for real-time particle-cascade exploration. Demo: QCD Jets Visualization.

PUBLICATIONS

- Laia, R.; Jorge, R.; Abreu, G. *Data-driven approach to model the influence of magnetic geometry in the confinement of fusion devices*. *Nuclear Fusion*, 66(1):016034, 2026. DOI: 10.1088/1741-4326/ae1e12.

TECHNICAL SKILLS

- **Programming:** Python, C#, C++, R, SQL.
- **ML/Data:** NumPy, pandas, scikit-learn, PyTorch, TensorFlow, LightGBM.

LANGUAGES

- Portuguese (Native), English (C1).

VOLUNTEERING

- **AEIST (Student Association):** Coordinated volunteers across multiple events, **streamlined** requests and **managed** the stationery working supplies.
- **ReFood: Recovered** food, **organized** routes to help multiple families.
- **Web Summit (Volunteer):** **Onboarded** speakers and attendees, **resolved** logistics issues at the conference.